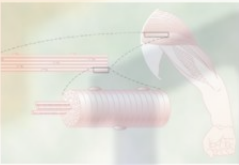


Contraction of Skeletal Muscle



About 40 per cent of the body is skeletal muscle, and perhaps another 10 per cent is smooth and cardiac muscle. Some of the same basic principles of contraction apply to all these different types of muscle. In this chapter, function of skeletal muscle is considered mainly; the specialized functions of smooth muscle are discussed in Chapter 8, and cardiac muscle is discussed in Chapter 9.

Physiologic Anatomy of Skeletal Muscle

Skeletal Muscle Fiber

Figure 6-1 shows the organization of skeletal muscle, demonstrating that all skeletal muscles are composed of numerous fibers ranging from 10 to 80 micrometers in diameter. Each of these fibers is made up of successively smaller subunits, also shown in Figure 6-1 and described in subsequent paragraphs.

In most skeletal muscles, each fiber extends the entire length of the muscle. Except for about 2 per cent of the fibers, each fiber is usually innervated by only one nerve ending, located near the middle of the fiber.

Sarcolemma. The sarcolemma is the cell membrane of the muscle fiber. The sarcolemma consists of a true cell membrane, called the *plasma membrane*, and an outer coat made up of a thin layer of polysaccharide material that contains numerous thin collagen fibrils. At each end of the muscle fiber, this surface layer of the sarcolemma fuses with a tendon fiber, and the tendon fibers in turn collect into bundles to form the muscle tendons that then insert into the bones.

Myofibrils; Actin and Myosin Filaments. Each muscle fiber contains several hundred to several thousand *myofibrils*, which are demonstrated by the many small open dots in the cross-sectional view of Figure 6-1C. Each myofibril (Figure 6-1D and E) is composed of about 1500 adjacent *myosin filaments* and 3000 *actin filaments*, which are large polymerized protein molecules that are responsible for the actual muscle contraction. These can be seen in longitudinal view in the electron micrograph of Figure 6-2 and are represented diagrammatically in Figure 6-1, parts E through L. The thick filaments in the diagrams are *myosin*, and the thin filaments are *actin*.

Note in Figure 6-1E that the myosin and actin filaments partially interdigitate and thus cause the myofibrils to have alternate light and dark bands, as illustrated in Figure 6-2. The light bands contain only actin filaments and are called *I bands* because they are *isotropic* to polarized light. The dark bands contain myosin filaments, as well as the ends of the actin filaments where they overlap the myosin, and are called *A bands* because they are *anisotropic* to polarized light. Note also the small projections from the sides of the myosin filaments in Figure 6-1E and L. These are *cross-bridges*. It is the interaction between these cross-bridges and the actin filaments that causes contraction.

Figure 6-1E also shows that the ends of the actin filaments are attached to a so-called *Z disc*. From this disc, these filaments extend in both directions to

SKELETAL MUSCLE

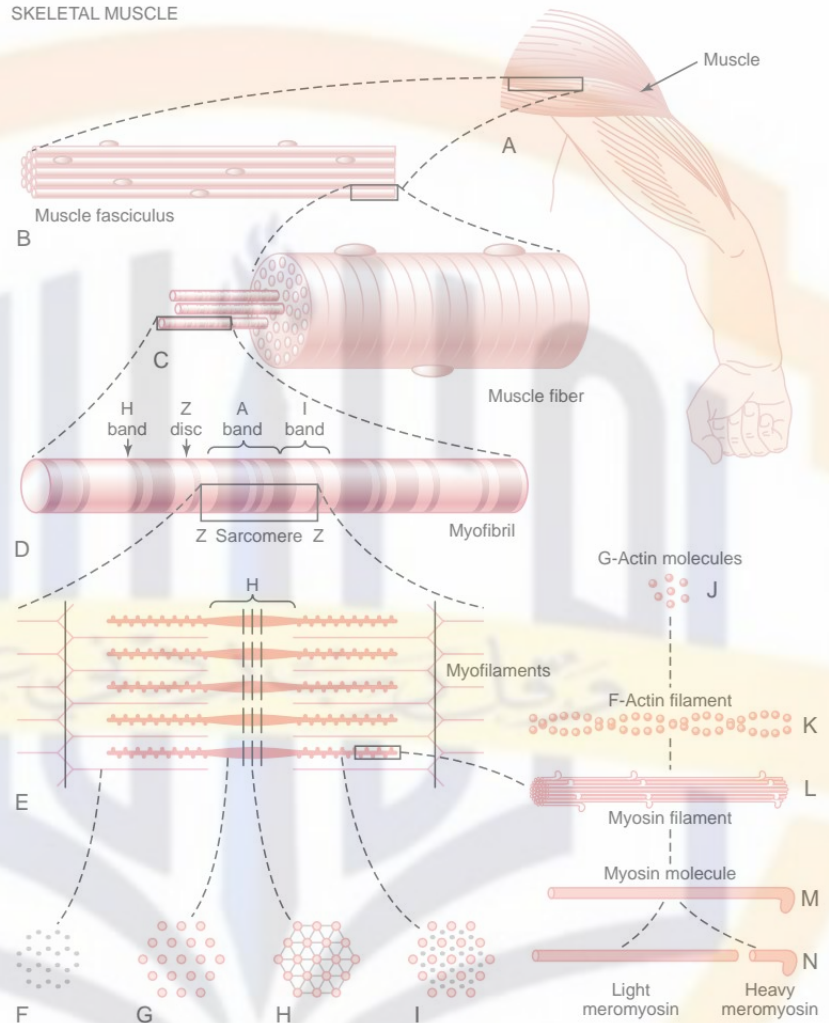


Figure 6-1

Organization of skeletal muscle, from the gross to the molecular level. *F*, *G*, *H*, and *I* are cross sections at the levels indicated. (Drawing by Sylvia Colard Keene. Modified from Fawcett DW: Bloom and Fawcett: A Textbook of Histology. Philadelphia: WB Saunders, 1986.)

interdigitate with the myosin filaments. The Z disc, which itself is composed of filamentous proteins different from the actin and myosin filaments, passes crosswise across the myofibril and also crosswise from myofibril to myofibril, attaching the myofibrils to one another all the way across the muscle fiber. Therefore, the entire muscle fiber has light and dark bands, as do the individual myofibrils. These bands give skeletal and cardiac muscle their striated appearance.

The portion of the myofibril (or of the whole muscle fiber) that lies between two successive Z discs is called a *sarcomere*. When the muscle fiber is contracted, as shown at the bottom of Figure 6-4, the length of the sarcomere is about 2 micrometers. At this length, the actin filaments completely overlap the myosin filaments, and the tips of the actin filaments are just

beginning to overlap one another. We will see later that, at this length, the muscle is capable of generating its greatest force of contraction.

What Keeps the Myosin and Actin Filaments in Place? Titin Filamentous Molecules. The side-by-side relationship between the myosin and actin filaments is difficult to maintain. This is achieved by a large number of filamentous molecules of a protein called *titin*. Each titin molecule has a molecular weight of about 3 million, which makes it one of the largest protein molecules in the body. Also, because it is filamentous, it is *very springy*. These springy titin molecules act as a framework that holds the myosin and actin filaments in place so that the contractile machinery of the sarcomere will work. There is reason to believe that the

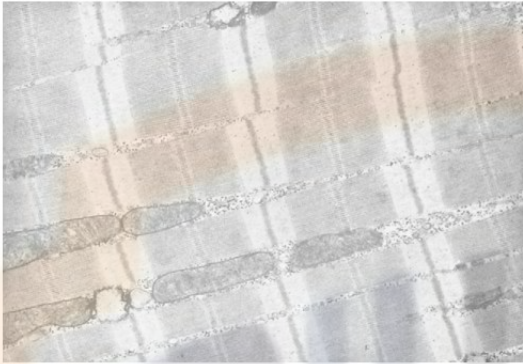


Figure 6-2

Electron micrograph of muscle myofibrils showing the detailed organization of actin and myosin filaments. Note the mitochondria lying between the myofibrils. (From Fawcett DW: *The Cell*. Philadelphia: WB Saunders, 1981.)

titin molecule itself acts as template for initial formation of portions of the contractile filaments of the sarcomere, especially the myosin filaments.

Sarcoplasm. The many myofibrils of each muscle fiber are suspended side by side in the muscle fiber. The spaces between the myofibrils are filled with intracellular fluid called sarcoplasm, containing large quantities of potassium, magnesium, and phosphate, plus multiple protein enzymes. Also present are tremendous numbers of *mitochondria* that lie parallel to the myofibrils. These supply the contracting myofibrils with large amounts of energy in the form of adenosine triphosphate (ATP) formed by the mitochondria.

Sarcoplasmic Reticulum. Also in the sarcoplasm surrounding the myofibrils of each muscle fiber is an extensive reticulum (Figure 6-3), called the *sarcoplasmic reticulum*. This reticulum has a special organization that is extremely important in controlling muscle contraction, as discussed in Chapter 7. The very rapidly contracting types of muscle fibers have especially extensive sarcoplasmic reticula.

General Mechanism of Muscle Contraction

The initiation and execution of muscle contraction occur in the following sequential steps.

1. An action potential travels along a motor nerve to its endings on muscle fibers.
2. At each ending, the nerve secretes a small amount of the neurotransmitter substance *acetylcholine*.
3. The acetylcholine acts on a local area of the muscle fiber membrane to open multiple “acetylcholine-gated” channels through protein molecules floating in the membrane.

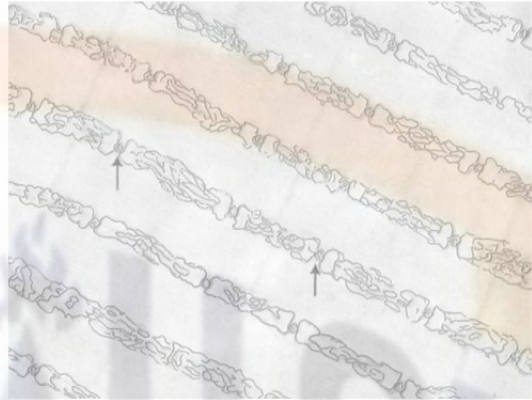


Figure 6-3

Sarcoplasmic reticulum in the extracellular spaces between the myofibrils, showing a longitudinal system paralleling the myofibrils. Also shown in cross section are T tubules (*arrows*) that lead to the exterior of the fiber membrane and are important for conducting the electrical signal into the center of the muscle fiber. (From Fawcett DW: *The Cell*. Philadelphia: WB Saunders, 1981.)

4. Opening of the acetylcholine-gated channels allows large quantities of sodium ions to diffuse to the interior of the muscle fiber membrane. This initiates an action potential at the membrane.
5. The action potential travels along the muscle fiber membrane in the same way that action potentials travel along nerve fiber membranes.
6. The action potential depolarizes the muscle membrane, and much of the action potential electricity flows through the center of the muscle fiber. Here it causes the sarcoplasmic reticulum to release large quantities of calcium ions that have been stored within this reticulum.
7. The calcium ions initiate attractive forces between the actin and myosin filaments, causing them to slide alongside each other, which is the contractile process.
8. After a fraction of a second, the calcium ions are pumped back into the sarcoplasmic reticulum by a Ca^{++} membrane pump, and they remain stored in the reticulum until a new muscle action potential comes along; this removal of calcium ions from the myofibrils causes the muscle contraction to cease.

We now describe the molecular machinery of the muscle contractile process.

Molecular Mechanism of Muscle Contraction

Sliding Filament Mechanism of Muscle Contraction. Figure 6-4 demonstrates the basic mechanism of muscle contraction. It shows the relaxed state of a sarcomere (top) and the contracted state (bottom). In the relaxed state, the ends of the actin filaments extending from two successive Z discs barely begin to overlap one another. Conversely, in the contracted state, these actin filaments have been pulled inward among the myosin

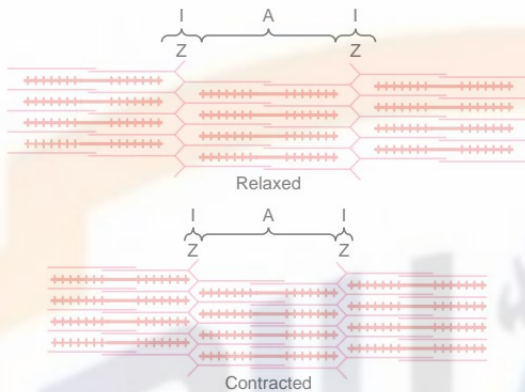


Figure 6-4

Relaxed and contracted states of a myofibril showing (top) sliding of the actin filaments (pink) into the spaces between the myosin filaments (red), and (bottom) pulling of the Z membranes toward each other.

filaments, so that their ends overlap one another to their maximum extent. Also, the Z discs have been pulled by the actin filaments up to the ends of the myosin filaments. Thus, muscle contraction occurs by a *sliding filament mechanism*.

But what causes the actin filaments to slide inward among the myosin filaments? This is caused by forces generated by interaction of the cross-bridges from the myosin filaments with the actin filaments. Under resting conditions, these forces are inactive, but when an action potential travels along the muscle fiber, this causes the sarcoplasmic reticulum to release large quantities of calcium ions that rapidly surround the myofibrils. The calcium ions in turn activate the forces between the myosin and actin filaments, and contraction begins. But energy is needed for the contractile process to proceed. This energy comes from high-energy bonds in the ATP molecule, which is degraded to adenosine diphosphate (ADP) to liberate the energy. In the next few sections, we describe what is known about the details of these molecular processes of contraction.

Molecular Characteristics of the Contractile Filaments

Myosin Filament. The myosin filament is composed of multiple myosin molecules, each having a molecular weight of about 480,000. Figure 6-5A shows an individual molecule; Figure 6-5B shows the organization of many molecules to form a myosin filament, as well as interaction of this filament on one side with the ends of two actin filaments.

The *myosin molecule* (see Figure 6-5A) is composed of six polypeptide chains—two *heavy chains*, each with a molecular weight of about 200,000, and four *light chains* with molecular weights of about 20,000 each.

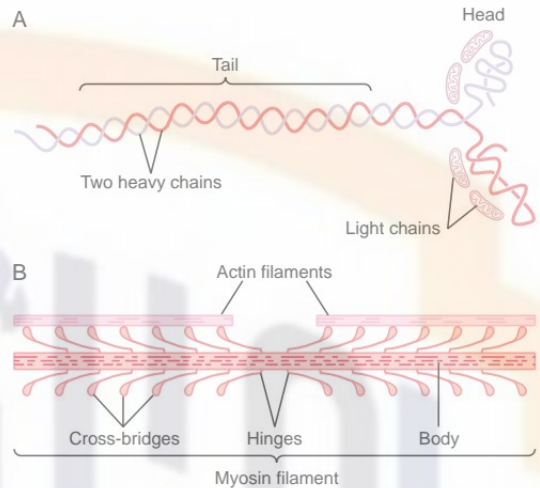


Figure 6-5

A, Myosin molecule. B, Combination of many myosin molecules to form a myosin filament. Also shown are thousands of myosin cross-bridges and interaction between the heads of the cross-bridges with adjacent actin filaments.

The two heavy chains wrap spirally around each other to form a double helix, which is called the *tail* of the myosin molecule. One end of each of these chains is folded bilaterally into a globular polypeptide structure called a *myosin head*. Thus, there are two free heads at one end of the double-helix myosin molecule. The four light chains are also part of the myosin head, two to each head. These light chains help control the function of the head during muscle contraction.

The *myosin filament* is made up of 200 or more individual myosin molecules. The central portion of one of these filaments is shown in Figure 6-5B, displaying the tails of the myosin molecules bundled together to form the *body* of the filament, while many heads of the molecules hang outward to the sides of the body. Also, part of the body of each myosin molecule hangs to the side along with the head, thus providing an *arm* that extends the head outward from the body, as shown in the figure. The protruding arms and heads together are called *cross-bridges*. Each cross-bridge is flexible at two points called *hinges*—one where the arm leaves the body of the myosin filament, and the other where the head attaches to the arm. The hinged arms allow the heads either to be extended far outward from the body of the myosin filament or to be brought close to the body. The hinged heads in turn participate in the actual contraction process, as discussed in the following sections.

The total length of each myosin filament is uniform, almost exactly 1.6 micrometers. Note, however, that there are no cross-bridge heads in the very center of the myosin filament for a distance of about 0.2 micrometer because the hinged arms extend away from the center.

Now, to complete the picture, the myosin filament itself is twisted so that each successive pair of cross-bridges is axially displaced from the previous pair by 120 degrees. This ensures that the cross-bridges extend in all directions around the filament.

ATPase Activity of the Myosin Head. Another feature of the myosin head that is essential for muscle contraction is that it functions as an *ATPase enzyme*. As explained later, this property allows the head to cleave ATP and to use the energy derived from the ATP's high-energy phosphate bond to energize the contraction process.

Actin Filament. The actin filament is also complex. It is composed of three protein components: *actin*, *tropomyosin*, and *troponin*.

The backbone of the actin filament is a double-stranded *F-actin protein molecule*, represented by the two lighter-colored strands in Figure 6-6. The two strands are wound in a helix in the same manner as the myosin molecule.

Each strand of the double F-actin helix is composed of polymerized *G-actin molecules*, each having a molecular weight of about 42,000. Attached to each one of the G-actin molecules is one molecule of ADP. It is believed that these ADP molecules are the active sites on the actin filaments with which the cross-bridges of the myosin filaments interact to cause muscle contraction. The active sites on the two F-actin strands of the double helix are staggered, giving one active site on the overall actin filament about every 2.7 nanometers.

Each actin filament is about 1 micrometer long. The bases of the actin filaments are inserted strongly into the Z discs; the ends of the filaments protrude in both directions to lie in the spaces between the myosin molecules, as shown in Figure 6-4.

Tropomyosin Molecules. The actin filament also contains another protein, *tropomyosin*. Each molecule of tropomyosin has a molecular weight of 70,000 and a length of 40 nanometers. These molecules are wrapped spirally around the sides of the F-actin helix. In the resting state, the tropomyosin molecules lie on top of the active sites of the actin strands, so that attraction

cannot occur between the actin and myosin filaments to cause contraction.

Troponin and Its Role in Muscle Contraction. Attached intermittently along the sides of the tropomyosin molecules are still other protein molecules called *troponin*. These are actually complexes of three loosely bound protein subunits, each of which plays a specific role in controlling muscle contraction. One of the subunits (troponin I) has a strong affinity for actin, another (troponin T) for tropomyosin, and a third (troponin C) for calcium ions. This complex is believed to attach the tropomyosin to the actin. The strong affinity of the troponin for calcium ions is believed to initiate the contraction process, as explained in the next section.

Interaction of One Myosin Filament, Two Actin Filaments, and Calcium Ions to Cause Contraction

Inhibition of the Actin Filament by the Troponin-Tropomyosin Complex; Activation by Calcium Ions. A pure actin filament without the presence of the troponin-tropomyosin complex (but in the presence of magnesium ions and ATP) binds instantly and strongly with the heads of the myosin molecules. Then, if the troponin-tropomyosin complex is added to the actin filament, the binding between myosin and actin does not take place. Therefore, it is believed that the active sites on the normal actin filament of the relaxed muscle are inhibited or physically covered by the troponin-tropomyosin complex. Consequently, the sites cannot attach to the heads of the myosin filaments to cause contraction. Before contraction can take place, the inhibitory effect of the troponin-tropomyosin complex must itself be inhibited.

This brings us to the role of the calcium ions. In the presence of large amounts of calcium ions, the inhibitory effect of the troponin-tropomyosin on the actin filaments is itself inhibited. The mechanism of this is not known, but one suggestion is the following: When calcium ions combine with troponin C, each molecule of which can bind strongly with up to four calcium ions, the troponin complex supposedly undergoes a conformational change that in some way tugs on the tropomyosin molecule and moves it deeper into the groove between the two actin strands. This "uncovers" the active sites of the actin, thus allowing these to attract the myosin cross-bridge heads and cause contraction to proceed. Although this is a hypothetical mechanism, it does emphasize that the normal relation between the troponin-tropomyosin complex and actin is altered by calcium ions, producing a new condition that leads to contraction.

Interaction Between the "Activated" Actin Filament and the Myosin Cross-Bridges—The "Walk-Along" Theory of Contraction. As soon as the actin filament becomes activated by the calcium ions, the heads of the cross-bridges from the myosin filaments become attracted to the active sites of the actin filament, and this, in some way, causes contraction to occur. Although the precise

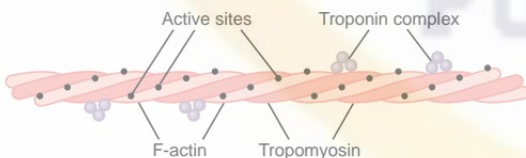


Figure 6-6

Actin filament, composed of two helical strands of *F-actin* molecules and two strands of *tropomyosin* molecules that fit in the grooves between the actin strands. Attached to one end of each tropomyosin molecule is a *troponin* complex that initiates contraction.

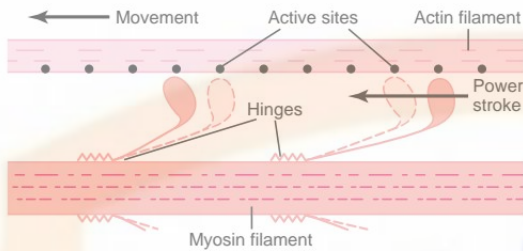


Figure 6-7

"Walk-along" mechanism for contraction of the muscle.

manner by which this interaction between the cross-bridges and the actin causes contraction is still partly theoretical, one hypothesis for which considerable evidence exists is the "walk-along" theory (or "ratchet" theory) of contraction.

Figure 6-7 demonstrates this postulated walk-along mechanism for contraction. The figure shows the heads of two cross-bridges attaching to and disengaging from active sites of an actin filament. It is postulated that when a head attaches to an active site, this attachment simultaneously causes profound changes in the intramolecular forces between the head and arm of its cross-bridge. The new alignment of forces causes the head to tilt toward the arm and to drag the actin filament along with it. This tilt of the head is called the *power stroke*. Then, immediately after tilting, the head automatically breaks away from the active site. Next, the head returns to its extended direction. In this position, it combines with a new active site farther down along the actin filament; then the head tilts again to cause a new power stroke, and the actin filament moves another step. Thus, the heads of the cross-bridges bend back and forth and step by step walk along the actin filament, pulling the ends of two successive actin filaments toward the center of the myosin filament.

Each one of the cross-bridges is believed to operate independently of all others, each attaching and pulling in a continuous repeated cycle. Therefore, the greater the number of cross-bridges in contact with the actin filament at any given time, the greater, theoretically, the force of contraction.

ATP as the Source of Energy for Contraction—Chemical Events in the Motion of the Myosin Heads. When a muscle contracts, work is performed and energy is required. Large amounts of ATP are cleaved to form ADP during the contraction process; the greater the amount of work performed by the muscle, the greater the amount of ATP that is cleaved, which is called the *Fenn effect*. The following sequence of events is believed to be the means by which this occurs:

1. Before contraction begins, the heads of the cross-bridges bind with ATP. The ATPase activity of the myosin head immediately cleaves the ATP but leaves the cleavage products, ADP plus phosphate

ion, bound to the head. In this state, the conformation of the head is such that it extends perpendicularly toward the actin filament but is not yet attached to the actin.

2. When the troponin-tropomyosin complex binds with calcium ions, active sites on the actin filament are uncovered, and the myosin heads then bind with these, as shown in Figure 6-7.
3. The bond between the head of the cross-bridge and the active site of the actin filament causes a conformational change in the head, prompting the head to tilt toward the arm of the cross-bridge. This provides the *power stroke* for pulling the actin filament. The energy that activates the power stroke is the energy already stored, like a "cocked" spring, by the conformational change that occurred in the head when the ATP molecule was cleaved earlier.
4. Once the head of the cross-bridge tilts, this allows release of the ADP and phosphate ion that were previously attached to the head. At the site of release of the ADP, a new molecule of ATP binds. This binding of new ATP causes detachment of the head from the actin.
5. After the head has detached from the actin, the new molecule of ATP is cleaved to begin the next cycle, leading to a new power stroke. That is, the energy again "cocks" the head back to its perpendicular condition, ready to begin the new power stroke cycle.
6. When the cocked head (with its stored energy derived from the cleaved ATP) binds with a new active site on the actin filament, it becomes uncocked and once again provides a new power stroke.

Thus, the process proceeds again and again until the actin filaments pull the Z membrane up against the ends of the myosin filaments or until the load on the muscle becomes too great for further pulling to occur.

Effect of Amount of Actin and Myosin Filament Overlap on Tension Developed by the Contracting Muscle

Figure 6-8 shows the effect of sarcomere length and amount of myosin-actin filament overlap on the active tension developed by a contracting muscle fiber. To the right, shown in black, are different degrees of overlap of the myosin and actin filaments at different sarcomere lengths. At point D on the diagram, the actin filament has pulled all the way out to the end of the myosin filament, with no actin-myosin overlap. At this point, the tension developed by the activated muscle is zero. Then, as the sarcomere shortens and the actin filament begins to overlap the myosin filament, the tension increases progressively until the sarcomere length decreases to about 2.2 micrometers. At this point, the actin filament has already overlapped all the cross-bridges of the myosin filament but has not yet reached the center of the myosin filament. With

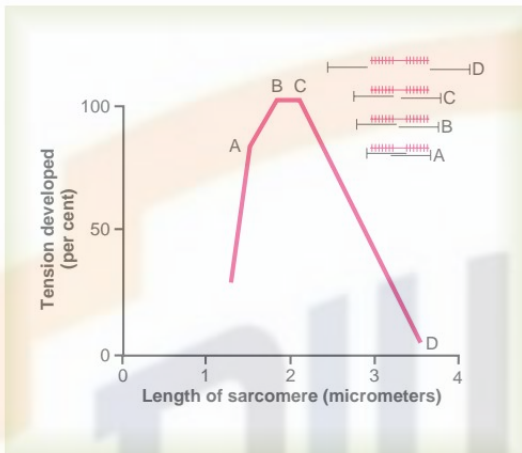


Figure 6-8

Length-tension diagram for a single fully contracted sarcomere, showing maximum strength of contraction when the sarcomere is 2.0 to 2.2 micrometers in length. At the upper right are the relative positions of the actin and myosin filaments at different sarcomere lengths from point A to point D. (Modified from Gordon AM, Huxley AF, Julian FJ: The length-tension diagram of single vertebrate striated muscle fibers. *J Physiol* 171:28P, 1964.)

further shortening, the sarcomere maintains full tension until point B is reached, at a sarcomere length of about 2 micrometers. At this point, the ends of the two actin filaments begin to overlap each other in addition to overlapping the myosin filaments. As the sarcomere length falls from 2 micrometers down to about 1.65 micrometers, at point A, the strength of contraction decreases rapidly. At this point, the two Z discs of the sarcomere abut the ends of the myosin filaments. Then, as contraction proceeds to still shorter sarcomere lengths, the ends of the myosin filaments are crumpled and, as shown in the figure, the strength of contraction approaches zero, but the entire muscle has now contracted to its shortest length.

Effect of Muscle Length on Force of Contraction in the Whole Intact Muscle. The top curve of Figure 6-9 is similar to that in Figure 6-8, but the curve in Figure 6-9 depicts tension of the intact, whole muscle rather than of a single muscle fiber. The whole muscle has a large amount of connective tissue in it; also, the sarcomeres in different parts of the muscle do not always contract the same amount. Therefore, the curve has somewhat different dimensions from those shown for the individual muscle fiber, but it exhibits the same general form for the slope in the normal range of contraction, as noted in Figure 6-9.

Note in Figure 6-9 that when the muscle is at its normal resting length, which is at a sarcomere length of about 2 micrometers, it contracts upon activation with the approximate maximum force of contraction. However, the increase in tension that occurs during contraction, called *active tension*, decreases as the

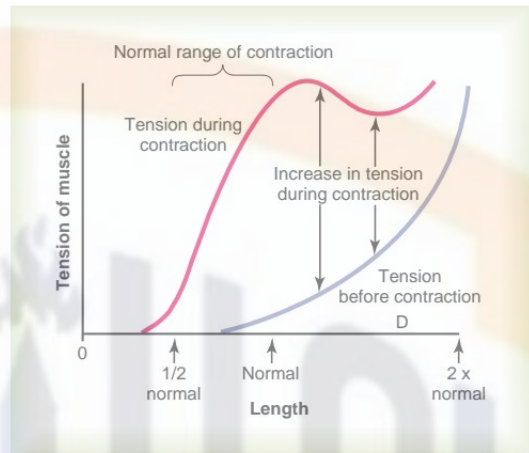


Figure 6-9

Relation of muscle length to tension in the muscle both before and during muscle contraction.

muscle is stretched beyond its normal length—that is, to a sarcomere length greater than about 2.2 micrometers. This is demonstrated by the decreased length of the arrow in the figure at greater than normal muscle length.

Relation of Velocity of Contraction to Load

A skeletal muscle contracts extremely rapidly when it contracts against no load—to a state of full contraction in about 0.1 second for the average muscle. When loads are applied, the velocity of contraction becomes progressively less as the load increases, as shown in Figure 6-10. That is, when the load has been increased to equal the maximum force that the muscle can exert, the velocity of contraction becomes zero and no contraction results, despite activation of the muscle fiber.

This decreasing velocity of contraction with load is caused by the fact that a load on a contracting muscle is a reverse force that opposes the contractile force caused by muscle contraction. Therefore, the net force that is available to cause velocity of shortening is correspondingly reduced.

Energetics of Muscle Contraction

Work Output During Muscle Contraction

When a muscle contracts against a load, it performs *work*. This means that *energy* is transferred from the muscle to the external load to lift an object to a greater height or to overcome resistance to movement.

In mathematical terms, work is defined by the following equation:

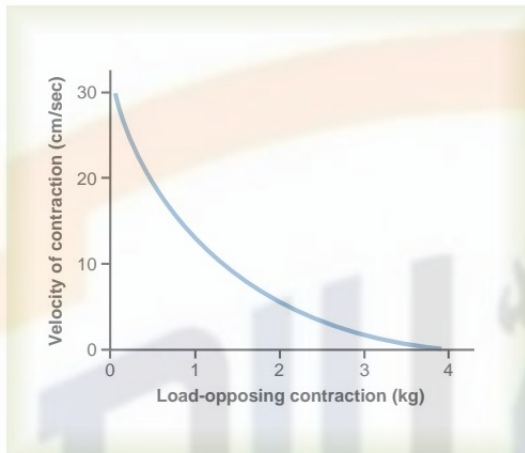


Figure 6-10

Relation of load to velocity of contraction in a skeletal muscle with a cross section of 1 square centimeter and a length of 8 centimeters.

$$W = L \times D$$

in which W is the work output, L is the load, and D is the distance of movement against the load. The energy required to perform the work is derived from the chemical reactions in the muscle cells during contraction, as described in the following sections.

Sources of Energy for Muscle Contraction

We have already seen that muscle contraction depends on energy supplied by ATP. Most of this energy is required to actuate the walk-along mechanism by which the cross-bridges pull the actin filaments, but small amounts are required for (1) pumping calcium ions from the sarcoplasm into the sarcoplasmic reticulum after the contraction is over, and (2) pumping sodium and potassium ions through the muscle fiber membrane to maintain appropriate ionic environment for propagation of muscle fiber action potentials.

The concentration of ATP in the muscle fiber, about 4 millimolar, is sufficient to maintain full contraction for only 1 to 2 seconds at most. The ATP is split to form ADP, which transfers energy from the ATP molecule to the contracting machinery of the muscle fiber. Then, as described in Chapter 2, the ADP is rephosphorylated to form new ATP within another fraction of a second, which allows the muscle to continue its contraction. There are several sources of the energy for this rephosphorylation.

The first source of energy that is used to reconstitute the ATP is the substance *phosphocreatine*, which carries a high-energy phosphate bond similar to the bonds of ATP. The high-energy phosphate bond of phosphocreatine has a slightly higher amount of free

energy than that of each ATP bond, as is discussed more fully in Chapters 67 and 72. Therefore, phosphocreatine is instantly cleaved, and its released energy causes bonding of a new phosphate ion to ADP to reconstitute the ATP. However, the total amount of phosphocreatine in the muscle fiber is also very little—only about five times as great as the ATP. Therefore, the combined energy of both the stored ATP and the phosphocreatine in the muscle is capable of causing maximal muscle contraction for only 5 to 8 seconds.

The second important source of energy, which is used to reconstitute both ATP and phosphocreatine, is “glycolysis” of *glycogen* previously stored in the muscle cells. Rapid enzymatic breakdown of the glycogen to pyruvic acid and lactic acid liberates energy that is used to convert ADP to ATP; the ATP can then be used directly to energize additional muscle contraction and also to re-form the stores of phosphocreatine.

The importance of this glycolysis mechanism is twofold. First, the glycolytic reactions can occur even in the absence of oxygen, so that muscle contraction can be sustained for many seconds and sometimes up to more than a minute, even when oxygen delivery from the blood is not available. Second, the rate of formation of ATP by the glycolytic process is about 2.5 times as rapid as ATP formation in response to cellular foodstuffs reacting with oxygen. However, so many end products of glycolysis accumulate in the muscle cells that glycolysis also loses its capability to sustain maximum muscle contraction after about 1 minute.

The third and final source of energy is *oxidative metabolism*. This means combining oxygen with the end products of glycolysis and with various other cellular foodstuffs to liberate ATP. More than 95 per cent of all energy used by the muscles for sustained, long-term contraction is derived from this source. The foodstuffs that are consumed are carbohydrates, fats, and protein. For extremely long-term maximal muscle activity—over a period of many hours—by far the greatest proportion of energy comes from fats, but for periods of 2 to 4 hours, as much as one half of the energy can come from stored carbohydrates.

The detailed mechanisms of these energetic processes are discussed in Chapters 67 through 72. In addition, the importance of the different mechanisms of energy release during performance of different sports is discussed in Chapter 84 on sports physiology.

Efficiency of Muscle Contraction. The efficiency of an engine or a motor is calculated as the percentage of energy input that is converted into work instead of heat. The percentage of the input energy to muscle (the chemical energy in nutrients) that can be converted into work, even under the best conditions, is less than 25 per cent, with the remainder becoming heat. The reason for this low efficiency is that about one half of the energy in foodstuffs is lost during the formation of ATP, and even then, only 40 to 45 per cent of the energy in the ATP itself can later be converted into work.

Maximum efficiency can be realized only when the muscle contracts at a moderate velocity. If the muscle contracts slowly or without any movement, small amounts of *maintenance heat* are released during contraction, even though little or no work is performed, thereby decreasing the conversion efficiency to as little as zero. Conversely, if contraction is too rapid, large proportions of the energy are used to overcome viscous friction within the muscle itself, and this, too, reduces the efficiency of contraction. Ordinarily, maximum efficiency is developed when the velocity of contraction is about 30 per cent of maximum.

Characteristics of Whole Muscle Contraction

Many features of muscle contraction can be demonstrated by eliciting single *muscle twitches*. This can be accomplished by instantaneous electrical excitation of the nerve to a muscle or by passing a short electrical stimulus through the muscle itself, giving rise to a single, sudden contraction lasting for a fraction of a second.

Isometric Versus Isotonic Contraction. Muscle contraction is said to be *isometric* when the muscle does not shorten during contraction and *isotonic* when it does shorten but the tension on the muscle remains constant throughout the contraction. Systems for recording the two types of muscle contraction are shown in Figure 6-11.

In the isometric system, the muscle contracts against a force transducer without decreasing the muscle length, as shown on the right in Figure 6-11. In the isotonic system, the muscle shortens against a fixed load; this is illustrated on the left in the figure, showing a muscle lifting a pan of weights. The characteristics of isotonic contraction depend on the load against which the muscle contracts, as well as the inertia of the load. However, the isometric system records strictly changes in force of muscle contraction itself. Therefore, the isometric system is most often used when comparing the functional characteristics of different muscle types.

Characteristics of Isometric Twitches Recorded from Different Muscles. The human body has many sizes of skeletal muscles—from the very small stapedius muscle in the middle ear, measuring only a few millimeters long and a millimeter or so in diameter, up to the very large

quadriceps muscle, a half million times as large as the stapedius. Further, the fibers may be as small as 10 micrometers in diameter or as large as 80 micrometers. Finally, the energetics of muscle contraction vary considerably from one muscle to another. Therefore, it is no wonder that the mechanical characteristics of muscle contraction differ among muscles.

Figure 6-12 shows records of isometric contractions of three types of skeletal muscle: an ocular muscle, which has a duration of *isometric* contraction of less than 1/40 second; the gastrocnemius muscle, which has a duration of contraction of about 1/15 second; and the soleus muscle, which has a duration of contraction of about 1/3 second. It is interesting that these durations of contraction are adapted to the functions of the respective muscles. Ocular movements must be extremely rapid to maintain fixation of the eyes on specific objects to provide accuracy of vision. The gastrocnemius muscle must contract moderately rapidly to provide sufficient velocity of limb movement for running and jumping, and the soleus muscle is concerned principally with slow contraction for continual, long-term support of the body against gravity.

Fast Versus Slow Muscle Fibers. As we discuss more fully in Chapter 84 on sports physiology, every muscle of the body is composed of a mixture of so-called *fast* and *slow* muscle fibers, with still other fibers gradated between these two extremes. The muscles that react rapidly are composed mainly of “fast” fibers with only small numbers of the slow variety. Conversely, the muscles that respond slowly but with prolonged contraction are composed mainly of “slow” fibers. The differences between these two types of fibers are as follows.

Fast Fibers. (1) Large fibers for great strength of contraction. (2) Extensive sarcoplasmic reticulum for rapid release of calcium ions to initiate contraction. (3) Large amounts of glycolytic enzymes for rapid release of energy by the glycolytic process. (4) Less extensive blood supply because oxidative metabolism is of

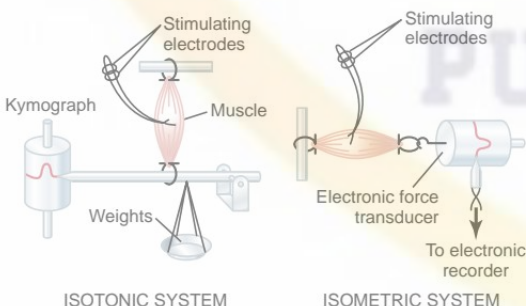


Figure 6-11

Isotonic and isometric systems for recording muscle contractions.

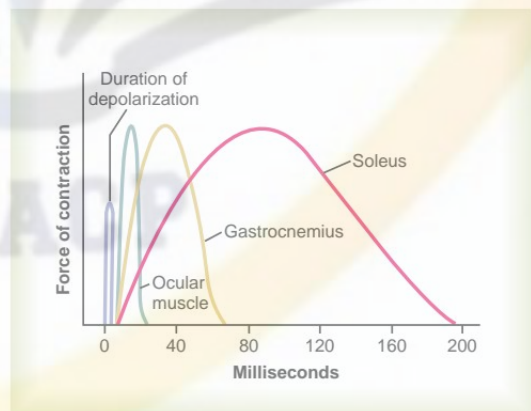


Figure 6-12

Duration of isometric contractions for different types of mammalian skeletal muscles, showing a latent period between the action potential (depolarization) and muscle contraction.

secondary importance. (5) Fewer mitochondria, also because oxidative metabolism is secondary.

Slow Fibers. (1) Smaller fibers. (2) Also innervated by smaller nerve fibers. (3) More extensive blood vessel system and capillaries to supply extra amounts of oxygen. (4) Greatly increased numbers of mitochondria, also to support high levels of oxidative metabolism. (5) Fibers contain large amounts of myoglobin, an iron-containing protein similar to hemoglobin in red blood cells. Myoglobin combines with oxygen and stores it until needed; this also greatly speeds oxygen transport to the mitochondria. The myoglobin gives the slow muscle a reddish appearance and the name *red muscle*, whereas a deficit of red myoglobin in fast muscle gives it the name *white muscle*.

Mechanics of Skeletal Muscle Contraction

Motor Unit. Each motoneuron that leaves the spinal cord innervates multiple muscle fibers, the number depending on the type of muscle. All the muscle fibers innervated by a single nerve fiber are called a *motor unit*. In general, small muscles that react rapidly and whose control must be exact have more nerve fibers for fewer muscle fibers (for instance, as few as two or three muscle fibers per motor unit in some of the laryngeal muscles). Conversely, large muscles that do not require fine control, such as the soleus muscle, may have several hundred muscle fibers in a motor unit. An average figure for all the muscles of the body is questionable, but a good guess would be about 80 to 100 muscle fibers to a motor unit.

The muscle fibers in each motor unit are not all bunched together in the muscle but overlap other motor units in microbundles of 3 to 15 fibers. This interdigitation allows the separate motor units to contract in support of one another rather than entirely as individual segments.

Muscle Contractions of Different Force—Force Summation. *Summation* means the adding together of individual twitch contractions to increase the intensity of overall muscle contraction. Summation occurs in two ways: (1) by increasing the number of motor units contracting simultaneously, which is called *multiple fiber summation*, and (2) by increasing the frequency of contraction, which is called *frequency summation* and can lead to *tetanzation*.

Multiple Fiber Summation. When the central nervous system sends a weak signal to contract a muscle, the smaller motor units of the muscle may be stimulated in preference to the larger motor units. Then, as the strength of the signal increases, larger and larger motor units begin to be excited as well, with the largest motor units often having as much as 50 times the contractile force of the smallest units. This is called the *size principle*. It is important, because it allows the gradations of muscle force during weak contraction to occur in small steps, whereas the steps become progressively greater when large amounts of force are required. The cause of this size principle is that the smaller motor units are driven by small motor nerve fibers, and the small motoneurons in the spinal cord are more excitable than the larger ones, so they naturally are excited first.

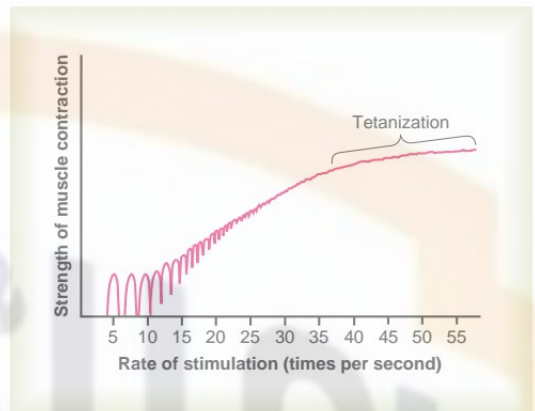


Figure 6-13

Frequency summation and tetanization.

Another important feature of multiple fiber summation is that the different motor units are driven asynchronously by the spinal cord, so that contraction alternates among motor units one after the other, thus providing smooth contraction even at low frequencies of nerve signals.

Frequency Summation and Tetanization. Figure 6-13 shows the principles of frequency summation and tetanization. To the left are displayed individual twitch contractions occurring one after another at low frequency of stimulation. Then, as the frequency increases, there comes a point where each new contraction occurs before the preceding one is over. As a result, the second contraction is added partially to the first, so that the total strength of contraction rises progressively with increasing frequency. When the frequency reaches a critical level, the successive contractions eventually become so rapid that they fuse together, and the whole muscle contraction appears to be completely smooth and continuous, as shown in the figure. This is called *tetanzation*. At a slightly higher frequency, the strength of contraction reaches its maximum, so that any additional increase in frequency beyond that point has no further effect in increasing contractile force. This occurs because enough calcium ions are maintained in the muscle sarcoplasm, even between action potentials, so that full contractile state is sustained without allowing any relaxation between the action potentials.

Maximum Strength of Contraction. The maximum strength of tetanic contraction of a muscle operating at a normal muscle length averages between 3 and 4 kilograms per square centimeter of muscle, or 50 pounds per square inch. Because a quadriceps muscle can have as much as 16 square inches of muscle belly, as much as 800 pounds of tension may be applied to the patellar tendon. Thus, one can readily understand how it is possible for muscles to pull their tendons out of their insertions in bone.

Changes in Muscle Strength at the Onset of Contraction—The Staircase Effect (Treppe). When a muscle begins to contract after a long period of rest, its initial strength of

contraction may be as little as one half its strength 10 to 50 muscle twitches later. That is, the strength of contraction increases to a plateau, a phenomenon called the *staircase effect*, or *treppe*.

Although all the possible causes of the staircase effect are not known, it is believed to be caused primarily by increasing calcium ions in the cytosol because of the release of more and more ions from the sarcoplasmic reticulum with each successive muscle action potential and failure of the sarcoplasm to recapture the ions immediately.

Skeletal Muscle Tone. Even when muscles are at rest, a certain amount of tautness usually remains. This is called *muscle tone*. Because normal skeletal muscle fibers do not contract without an action potential to stimulate the fibers, skeletal muscle tone results entirely from a low rate of nerve impulses coming from the spinal cord. These, in turn, are controlled partly by signals transmitted from the brain to the appropriate spinal cord anterior motoneurons and partly by signals that originate in *muscle spindles* located in the muscle itself. Both of these are discussed in relation to muscle spindle and spinal cord function in Chapter 54.

Muscle Fatigue. Prolonged and strong contraction of a muscle leads to the well-known state of muscle fatigue. Studies in athletes have shown that muscle fatigue increases in almost direct proportion to the rate of depletion of muscle glycogen. Therefore, fatigue results mainly from inability of the contractile and metabolic processes of the muscle fibers to continue supplying the same work output. However, experiments have also shown that transmission of the nerve signal through the neuromuscular junction, which is discussed in Chapter 7, can diminish at least a small amount after intense prolonged muscle activity, thus further diminishing muscle contraction. Interruption of blood flow through a contracting muscle leads to almost complete muscle fatigue within 1 or 2 minutes because of the loss of nutrient supply, especially loss of oxygen.

Lever Systems of the Body. Muscles operate by applying tension to their points of insertion into bones, and the bones in turn form various types of lever systems. Figure 6-14 shows the lever system activated by the biceps muscle to lift the forearm. If we assume that a large



Figure 6-14

Lever system activated by the biceps muscle.

biceps muscle has a cross-sectional area of 6 square inches, the maximum force of contraction would be about 300 pounds. When the forearm is at right angles with the upper arm, the tendon attachment of the biceps is about 2 inches anterior to the fulcrum at the elbow, and the total length of the forearm lever is about 14 inches. Therefore, the amount of lifting power of the biceps at the hand would be only one seventh of the 300 pounds of muscle force, or about 43 pounds. When the arm is fully extended, the attachment of the biceps is much less than 2 inches anterior to the fulcrum, and the force with which the hand can be brought forward is also much less than 43 pounds.

In short, an analysis of the lever systems of the body depends on knowledge of (1) the point of muscle insertion, (2) its distance from the fulcrum of the lever, (3) the length of the lever arm, and (4) the position of the lever. Many types of movement are required in the body, some of which need great strength and others of which need large distances of movement. For this reason, there are many different types of muscle; some are long and contract a long distance, and some are short but have large cross-sectional areas and can provide extreme strength of contraction over short distances. The study of different types of muscles, lever systems, and their movements is called *kinesiology* and is an important scientific component of human physioanatomy.

“Positioning” of a Body Part by Contraction of Agonist and Antagonist Muscles on Opposite Sides of a Joint—“Coactivation” of Antagonist Muscles. Virtually all body movements are caused by simultaneous contraction of agonist and antagonist muscles on opposite sides of joints. This is called *coactivation* of the agonist and antagonist muscles, and it is controlled by the motor control centers of the brain and spinal cord.

The position of each separate part of the body, such as an arm or a leg, is determined by the relative degrees of contraction of the agonist and antagonist sets of muscles. For instance, let us assume that an arm or a leg is to be placed in a midrange position. To achieve this, agonist and antagonist muscles are excited about equally. Remember that an elongated muscle contracts with more force than a shortened muscle, which was demonstrated in Figure 6-9, showing maximum strength of contraction at full functional muscle length and almost no strength of contraction at half normal length. Therefore, the elongated muscle on one side of a joint can contract with far greater force than the shorter muscle on the opposite side. As an arm or leg moves toward its midposition, the strength of the longer muscle decreases, whereas the strength of the shorter muscle increases until the two strengths equal each other. At this point, movement of the arm or leg stops. Thus, by varying the ratios of the degree of activation of the agonist and antagonist muscles, the nervous system directs the positioning of the arm or leg.

We learn in Chapter 54 that the motor nervous system has additional important mechanisms to compensate for different muscle loads when directing this positioning process.

Remodeling of Muscle to Match Function

All the muscles of the body are continually being remodeled to match the functions that are required of

them. Their diameters are altered, their lengths are altered, their strengths are altered, their vascular supplies are altered, and even the types of muscle fibers are altered at least slightly. This remodeling process is often quite rapid, within a few weeks. Indeed, experiments in animals have shown that muscle contractile proteins in some smaller, more active muscles can be replaced in as little as 2 weeks.

Muscle Hypertrophy and Muscle Atrophy. When the total mass of a muscle increases, this is called *muscle hypertrophy*. When it decreases, the process is called *muscle atrophy*.

Virtually all muscle hypertrophy results from an increase in the number of actin and myosin filaments in each muscle fiber, causing enlargement of the individual muscle fibers; this is called simply *fiber hypertrophy*. Hypertrophy occurs to a much greater extent when the muscle is loaded during the contractile process. Only a few strong contractions each day are required to cause significant hypertrophy within 6 to 10 weeks.

The manner in which forceful contraction leads to hypertrophy is not known. It is known, however, that the rate of synthesis of muscle contractile proteins is far greater when hypertrophy is developing, leading also to progressively greater numbers of both actin and myosin filaments in the myofibrils, often increasing as much as 50 per cent. In turn, some of the myofibrils themselves have been observed to split within hypertrophying muscle to form new myofibrils, but how important this is in usual muscle hypertrophy is still unknown.

Along with the increasing size of myofibrils, the enzyme systems that provide energy also increase. This is especially true of the enzymes for glycolysis, allowing rapid supply of energy during short-term forceful muscle contraction.

When a muscle remains unused for many weeks, the rate of decay of the contractile proteins is more rapid than the rate of replacement. Therefore, muscle atrophy occurs.

Adjustment of Muscle Length. Another type of hypertrophy occurs when muscles are stretched to greater than normal length. This causes new sarcomeres to be added at the ends of the muscle fibers, where they attach to the tendons. In fact, new sarcomeres can be added as rapidly as several per minute in newly developing muscle, illustrating the rapidity of this type of hypertrophy.

Conversely, when a muscle continually remains shortened to less than its normal length, sarcomeres at the ends of the muscle fibers can actually disappear. It is by these processes that muscles are continually remodeled to have the appropriate length for proper muscle contraction.

Hyperplasia of Muscle Fibers. Under rare conditions of extreme muscle force generation, the actual number of muscle fibers has been observed to increase (but only by a few percentage points), in addition to the fiber hypertrophy process. This increase in fiber number is called *fiber hyperplasia*. When it does occur, the mechanism is linear splitting of previously enlarged fibers.

Effects of Muscle Denervation. When a muscle loses its nerve supply, it no longer receives the contractile signals that are required to maintain normal muscle size. Therefore, atrophy begins almost immediately. After

about 2 months, degenerative changes also begin to appear in the muscle fibers themselves. If the nerve supply to the muscle grows back rapidly, full return of function can occur in as little as 3 months, but from that time onward, the capability of functional return becomes less and less, with no further return of function after 1 to 2 years.

In the final stage of denervation atrophy, most of the muscle fibers are destroyed and replaced by fibrous and fatty tissue. The fibers that do remain are composed of a long cell membrane with a lineup of muscle cell nuclei but with few or no contractile properties and little or no capability of regenerating myofibrils if a nerve does regrow.

The fibrous tissue that replaces the muscle fibers during denervation atrophy also has a tendency to continue shortening for many months, which is called *contracture*. Therefore, one of the most important problems in the practice of physical therapy is to keep atrophying muscles from developing debilitating and disfiguring contractures. This is achieved by daily stretching of the muscles or use of appliances that keep the muscles stretched during the atrophying process.

Recovery of Muscle Contraction in Poliomyelitis: Development of Macromotor Units. When some but not all nerve fibers to a muscle are destroyed, as commonly occurs in poliomyelitis, the remaining nerve fibers branch off to form new axons that then innervate many of the paralyzed muscle fibers. This causes large motor units called *macromotor units*, which can contain as many as five times the normal number of muscle fibers for each motoneuron coming from the spinal cord. This decreases the fineness of control one has over the muscles but does allow the muscles to regain varying degrees of strength.

Rigor Mortis

Several hours after death, all the muscles of the body go into a state of *contracture* called “rigor mortis”; that is, the muscles contract and become rigid, even without action potentials. This rigidity results from loss of all the ATP, which is required to cause separation of the cross-bridges from the actin filaments during the relaxation process. The muscles remain in rigor until the muscle proteins deteriorate about 15 to 25 hours later, which presumably results from autolysis caused by enzymes released from lysosomes. All these events occur more rapidly at higher temperatures.

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